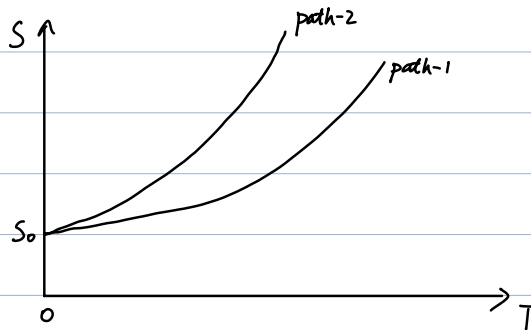


The third law of thermodynamics

Nernst statement:

$$\lim_{T \rightarrow 0} \Delta S = 0 \quad \text{for isothermal process}$$

i.e. $\forall x, \lim_{T \rightarrow 0} |S(T, x) - S(T, x + \Delta x)| \rightarrow 0$ where x is a state variable.



(x can be different for path-1 & 2)

$$T \rightarrow 0 \\ \Rightarrow S \rightarrow S_0 \quad \text{a constant}$$

Example, ideal gas (bad example)

$$S(T, V_1) - S(T, V_2) = nR \ln \frac{V_1}{V_2} \quad \text{independent of } T$$

Q: 3rd law violated?

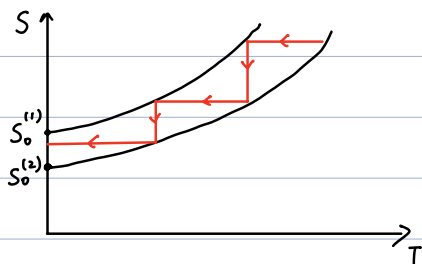
A: No. "Ideal gas model" too "ideal".

Unattainability principle:

(recall previous classes)

It is impossible for any process, no matter how idealized, to reduce the entropy of a system to its absolute-zero value in a finite number of operations.

If 3rd law violated (S at $T=0$ NOT well defined)



Reach $T=0$ in finite steps

(unattainability principle broken)

If 3rd law respected ($S = S_0$ at $T = 0$)



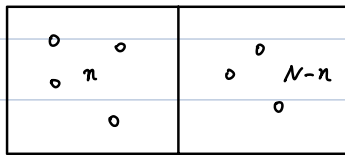
Can NOT reach $T=0$ in finite steps
(unattainability principle respected)

Statistical view of entropy

$$S = k_B \ln W$$

$\nwarrow$ multiplicity
 $\swarrow$ Boltzmann constant

1877 Ludwig Boltzmann



W : # of ways to distribute N particles
into two boxes with n in left
 $\Rightarrow W = \frac{N!}{n!(N-n)!}$

Example. Adiabatic free expansion of gas

$$V_f / V_i = 2 \Rightarrow \Delta S = nR \ln 2$$

$$W_i = \frac{N!}{N! 0!} = 1$$

$$\Rightarrow S_i = k_B \ln 1 = 0$$

$$W_f = \frac{N!}{(N/2)!(N/2)!}$$

$$\Rightarrow S_f = k_B \ln N! - 2k_B \ln (N/2)!$$

Stirling's approximation: $\ln N! \approx N \ln N - N$ for $N \gg 1$.

$$\Rightarrow S_f \approx k_B (N \ln N - N) - 2k_B \left(\frac{N}{2} \ln \frac{N}{2} - \frac{N}{2} \right)$$

$$= Nk_B \ln N - Nk_B \ln \frac{N}{2}$$

$$= Nk_B \ln 2$$

$$= nR \ln 2$$

Reminder: $R = N_A k_B$

$$\Rightarrow \Delta S = S_f - S_i \approx nR \ln 2$$